Mazen Elnaghy

AI Engineer
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CAREER OBJECTIVE

AI Engineer at junior level, training and deploying machine learning and computer vision models in Python and C++, from dataset annotation to real-time inference on edge devices such as the Raspberry Pi. Ranked 1st in my AI & Machine Learning Engineering department, with a data engineering foundation from the Digital Egypt Pioneers Initiative (DEPI). I build models that work on real inputs, not just on clean sample data.

EDUCATION

B.Sc. in Artificial Intelligence Engineering 2023 – 2028 (Expected)
East Port Said National University, Port Said, Egypt
GPA: 3.97 / 4.0 — Ranked 1st in the AI & Machine Learning Engineering department (2024 – 2025)
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Robotics

Scholarships
Data Engineering Track — Digital Egypt Pioneers Initiative (DEPI), MCIT June 2026 – February 2027
Competitive MCIT scholarship. SQL, ETL pipelines, data modelling, data warehousing, cloud data platforms.
- Built ETL and ELT pipelines in Python and SQL that turn raw source data into clean, analysis-ready datasets.

COURSES AND CERTIFICATES

Machine Learning Program — National Telecommunication Institute (NTI) January – April 2024
Python Programming Course — Codezilla June – September 2024
Huawei Technical Training Programs — Artificial Intelligence & ICT tracks

SKILLS

Programming Languages: Python (proficient), C++ (proficient), SQL (proficient), MicroPython (basic understanding)
Machine Learning: Machine learning, deep learning, neural networks, natural language processing, large language models
Computer Vision: YOLO (Ultralytics), OpenCV, object detection, image processing, real-time inference, dataset annotation
ML and Data Libraries: TensorFlow, Scikit-learn (familiar with), pandas (proficient), NumPy (proficient)
Edge and Robotics: Raspberry Pi, edge deployment, ROS 2
Data Engineering: ETL and ELT pipelines, DuckDB, dimensional modelling (star schema), data quality and validation
Tools and Platforms: Git, GitHub Actions (CI/CD), Linux, Jupyter Notebook, Anaconda, PyCharm, Visual Studio Code
Languages: Arabic (native), English (professional working proficiency, C1)

PROJECTS

AI Dog Detection and Deterrent System — github.com/mazenelnaghy-code/dog-detection
Problem: stray dogs near kindergartens and residential entrances put people at risk, with no automatic way to detect and deter them.
- Deployed YOLO on a Raspberry Pi for real-time edge inference, with a danger-zone rule and camera-based distance estimation.
- Raised detection on 236 Egyptian street-dog photos from 72% to 97% (4% false alarms on children) by remapping look-alike classes.
- Drove an ultrasonic repeller via a relay with burst/cooldown control and a safety watchdog; web dashboard, 100+ tests, CI.

Tools: Python, YOLO, OpenCV, Raspberry Pi, NCNN, Flask, GitHub Actions

Real-Time Orange Ball Detection System — github.com/mazenelnaghy-code/orange-ball-detector
Problem: a specific object had to be detected and tracked reliably in a live camera feed under changing lighting and backgrounds.
- Built the full pipeline end to end: dataset collection and annotation, custom YOLO training and evaluation, and real-time inference.
- Implemented real-time tracking in Python with OpenCV, tuning the model to hold accuracy under varying lighting and backgrounds.

Tools: Python, YOLO, Ultralytics, OpenCV, dataset annotation

Egypt Air Quality Data Warehouse — github.com/mazenelnaghy-code/egypt-air-quality
Problem: air quality and weather readings for Egyptian cities exist only as live API calls, with no clean historical record to analyse.
- Built an automated Python pipeline that collects hourly air quality and weather data for 8 Egyptian cities from the Open-Meteo API into a DuckDB star schema, orchestrated on a 6-hourly schedule with GitHub Actions.
- Implemented 12 data quality assertions that block publication whenever a check fails, and made rebuilds idempotent by de-duplicating overlapping API windows.

Tools: Python, SQL, DuckDB, ETL and ELT pipelines, data quality validation, GitHub Actions
Live dashboard: mazenelnaghy-code.github.io/egypt-air-quality

EXPERIENCE

Programming Intern August 2026
Suez Canal Authority, Port Said, Egypt
- Completed a one-month intensive software development programme inside a live engineering environment.
- Developed and debugged Python and C++ applications, applying object-oriented design to real operational requirements.
- Designed UI/UX layouts and user flows that made technical tools clear and easy to operate for non-technical staff.
- Worked with engineers and technical staff to turn operational requirements into documented, working software tools.

HONOURS AND AWARDS

- Certificate of Excellence — ranked 1st in AI & Machine Learning Engineering, East Port Said National University (2024 – 2025).
- Certificate of Recognition — Outstanding Engineering Project, EAST Engineering Expo (2026).
- Digitopia 2025 — participant, national digital innovation competition.